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Mechanisms to Address Conceptual Gaps in Deterministic Wave Theory

1. Global Phase-Ordered Narrowband Carrier Sector

In a nonlinear elastic field with geometric nonlinearity, a globally phase-coherent oscillation (a single narrowband "carrier" frequency across the system) could emerge via self-organization akin to a **condensate** or **limit-cycle oscillation**. For example, in a Bose–Einstein condensate (or a driven exciton-polariton condensate), many modes lock into **one coherent phase** – essentially all oscillators fall into a single narrowband frequency with global phase order [1]. Such phase ordering can arise in driven-dissipative systems (like lasers) where above a threshold the system **spontaneously synchronizes** into a single-frequency oscillation (Haken's **laser self-organization** analogy). In a continuous nonlinear medium, **mode competition** and nonlinear frequency shifting could similarly favor one persistent carrier mode: energy fed into the system (or conserved in a nearly lossless medium) may cascade into a dominant mode that **phase-locks** the field globally.

- **Driven-Dissipative Condensates:** If the elastic field is pumped (energy input) and has slight damping, it can settle into a **steady oscillation**. This is analogous to how a laser's gain and loss balance produces a steady narrowband beam, or how a parametrically driven medium can oscillate at a selected frequency. The **global phase coherence** in such systems is maintained by continuous driving and nonlinear saturation (which clamps amplitude) [1]. In a mechanical analog, one could imagine an elastic medium under periodic driving so that one eigenmode (or a band of modes) dominates and phase-aligns across space. This resembles a **time-crystalline** state (periodic in time) stabilized by energy flow.

- **Floquet and Parametric Mechanisms:** Even without external pumping, nonlinear feedback can sustain a carrier. If the elastic theory permits a **limit cycle** solution (through a Hopf bifurcation), the whole field may oscillate in unison at a self-selected narrow frequency. *Floquet systems* offer an analog: a strong periodic modulation of system parameters can entrain the field to a single frequency (the Floquet frequency). In a nonlinear lattice, for instance, parametric resonance can amplify a specific mode. The phenomenon of **discrete breathers** in nonlinear lattices is illustrative: these are spatially localized, *time-periodic* oscillations that persist at a fixed frequency [2]. A large-scale analog would be a **nonlinear normal mode** of the continuum – effectively, a **collective oscillation mode** where geometric nonlinearity prevents decay into turbulence. Geometric nonlinearity (strain-dependent stiffness) can provide the self-tuning: as the amplitude grows, the frequency shifts to avoid resonances with other modes (a form of nonlinear detuning), thereby **protecting a single narrowband** oscillation.

- **Stability and Maintenance:** Once established, such a global carrier could be maintained by either **conservative trapping** or slight pumping to counteract dissipation. If the elastic medium has very low damping, a large-scale coherent oscillation can persist for cosmologically long times (similar to how a superfluid supports a persistent current). If some damping is present, an external or internal energy source (e.g. an ambient "zero-point" drive or continual input from a parallel domain) would be needed to sustain the oscillation. Notably, **phase-ordered oscillations** are seen in extended nonlinear systems from plasmas to fluid flows when a symmetry is broken and one mode dominates (sometimes called **self-organized oscillations**).

Thus, by analogy to lasers and condensates, one can posit that the nonlinear elastic spacetime may undergo a **self-organized synchronization** into a narrowband mode. This addresses how a stable carrier wave might emerge: it could be a **metastable collective mode** of the nonlinear field, similar to a macroscopic condensate, that remains phase-coherent across space and time.

2. Emergent Gauge Connection from Local Mode Structure

In an inhomogeneous or modulated elastic substrate, the **local polarization eigenmodes** of wave motion can vary with position – this is directly analogous to having an internal "fiber bundle" over space. A slowly-varying local basis of polarization (or other internal degrees of freedom) will induce a **Berry connection** (geometric phase) when waves traverse the medium. This can give rise to an **effective gauge field** experienced by the envelope of the wave. In other words, if the preferred polarization or phase of the carrier rotates adiabatically in space, the envelope's phase picks up an extra path-dependent phase factor just like a charged particle in a gauge potential. For example, recent work in photonics demonstrates that spatially varying anisotropy in a medium yields an emergent **non-Abelian gauge field** for light [3] [4]. The medium's anisotropy acts as a position-dependent "Hamiltonian" for polarization states; as light adiabatically follows the local eigen-polarization, it acquires a Berry-Wilczek–Zee phase analogous to a **Wilson loop** in gauge theory [5] [6]. This is essentially a **Wilczek–Zee connection** (non-Abelian Berry phase) arising from a degenerate set of modes.

Under what conditions would this behave like a dynamical gauge field (with its own field curvature obeying Maxwell/Yang–Mills equations)? We require that the **inhomogeneity is itself dynamic** or can transmit disturbances. In an elastic continuum, parameters like local stiffness or orientation can change slowly – these changes propagate as another mode (e.g. elastic modulations or defects moving). If those "parameters" are promoted to fields, then their slow dynamics could be described by an action containing a **field strength**. A concrete analogue is found in **spin liquid** or **electromagnetic metamaterials**: in quantum spin ice, the orientation of spins leads to an emergent U(1) gauge field with real dynamic photons – the long-wavelength excitations are **Maxwell-like** and mediate a Coulomb potential between emergent charges (monopoles) [7] [8]. Similarly, in a continuum, if the local eigenmode basis evolves in time, one can define an **effective gauge field** A_μ^a (for an internal index a labeling modes) from the spatial gradients of the mode basis. In the adiabatic limit, the wave's envelope equation would contain a covariant derivative $D_\mu = \partial_\mu + iA_\mu$ acting on the multi-component envelope [9] [10]. The **gauge curvature** $F_{\mu\nu}$ arises from the non-commutativity of $D_\mu D_\nu$ – essentially the Berry curvature in real space. This curvature can deflect or split wave packets analogously to the Lorentz force (indeed, experiments show *optical Zitterbewegung* and spin Hall effects of light due to synthetic gauge fields [11] [9]).

To obey a Maxwell- or Yang–Mills-like equation, the gauge field needs a kinetic term. In an effective field theory obtained by coarse-graining the elastic medium, such a term can appear if changes in the polarization field cost energy (i.e. the medium resists rapid twists in polarization). For small modulations, one can expand the elastic energy in gradients of the local polarization basis – this yields a term like $(\nabla A)^2$ in the effective Lagrangian, which is analogous to a field strength squared ($F_{\mu\nu}F^{\mu\nu}$) [12] [13]. Under suitable approximations (e.g. weak inhomogeneity, adiabatic following of modes), this gauge field will satisfy **homogeneous Maxwell equations** (from the Bianchi identity for Berry curvature) and **inhomogeneous equations** sourced by defects in the polarization field (analogous to "electric" charges, which could be localized mode-mixing sources). In essence, the elastic medium's internal structure provides a "fiber bundle," and if that structure can oscillate or propagate, the emergent Berry connection becomes a **dynamical gauge field**. Literature in transformation optics and metamaterials has already engineered **synthetic Abelian gauge fields** in real space [6] [14], and even proposed non-Abelian gauge potentials for two polarizations of light [3] [15]. These are typically background fields, but one can imagine the substrate's properties being **highly**

flexible (elastic) so that those gauge fields respond to the presence of energy/momentum – thereby obeying an effective Yang–Mills equation. Under a separation of scales (carrier vs. envelope), the leading order emergent gauge fields would be *source-free* (parallel transport around a loop yields the Berry phase), and higher-order corrections would incorporate coupling to "charges" (e.g. envelope polarization defects).

In summary, a **Berry-phase-induced gauge field** can be derived from spatially varying polarization eigenmodes of an inhomogeneous medium. If the medium's parameters are treated as fields with their own dynamics (e.g. through a Born-Oppenheimer separation or multi-scale expansion), the gauge connection gains kinetic energy and curvature terms. In the long-wavelength effective action, this can produce terms analogous to $\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, and thus the **emergent gauge field evolves according to Maxwell/Yang–Mills equations** (at least approximately, within the regime where the envelope description holds). This mechanism is conceptually similar to how **Wilczek–Zee non-Abelian phases** appear in multi-component quantum systems, or how **elastic deformations in a moving medium** can mimic electromagnetic fields (e.g. "frame dragging" effects in solids acting like gauge potentials). It provides a concrete route to incorporate gauge connections into the deterministic wave framework via the medium's internal mode structure [3] [9].

3. Envelope Modulation and Broadband Electromagnetic-Like Phenomena

A key idea of the deterministic wave theory is that *low-frequency phenomena (electromagnetism)* arise as **modulations of a high-frequency carrier** field. The mathematics of multi-scale analysis supports this: if $u(x,t)$ is the full displacement field, one can write it as a rapidly oscillating carrier with a slowly varying envelope, e.g. $u(x,t) = A(X,T)e^{i\theta(x,t)}$ with θ a fast phase $k_0 x - \omega_0 t$ [16]. The envelope $A(X,T)$ (with X,T denoting slow spatial and temporal scales) contains **broadband content** even though the carrier is narrowband. How can this produce electromagnetic-like behavior? Essentially, the *envelope* obeys a wave equation (or a set of coupled equations) that is not limited to the narrow frequency of the carrier. All the richness of electromagnetic phenomena – static Coulomb fields, propagating waves of various frequencies, etc. – can emerge in the envelope dynamics, riding on the back of the high-frequency carrier.

- **Broadband from Narrowband:** When you modulate a single-frequency wave, you generate sidebands. In a continuous envelope description, the envelope can vary arbitrarily slowly or quickly compared to the carrier oscillation. For instance, consider a pure tone $e^{i\omega_0 t}$ whose amplitude is modulated at a much lower frequency Ω: the total signal contains frequencies $\omega_0 \pm \Omega$. By layering many Fourier components in the envelope, one can synthesize essentially any **broadband signal** on the narrowband carrier. In the context of the theory, the carrier might oscillate at some ultra-high "microscopic" frequency (perhaps near a cutoff or intrinsic scale), while the envelope can represent phenomena at all lower frequencies (from DC up to some cutoff below the carrier). Therefore, **electromagnetic fields** – which span a broad spectrum from static (0 Hz) up to optical frequencies – could correspond to patterns on the carrier wave. A static Coulomb potential, for example, would be represented by a *slowly varying envelope amplitude* (essentially a DC spatial variation) on the fast oscillation. In a multi-scale expansion of a nonlinear wave equation, the envelope often satisfies a wave equation similar to Maxwell's equations. Indeed, the deterministic elasticity model cited posits that after coarse-graining the fast oscillations, the envelope $\Psi(x,t)$ **"obeys an effective Schrödinger-like equation"**, reproducing interference and wave dynamics [17] [18]. By extension, if multiple envelope components are considered (e.g. representing electric and magnetic field components), one can obtain **effective Maxwell equations for the envelope**.

- **Effective Charges and Coulomb Law:** In a classical wave system, what would correspond to an electric charge? A charge is a source of an electric field – in this analogy, a "source" would be any localized disturbance of the envelope that causes a $1/r$-type envelope profile. For example, a localized oscillation of the carrier with a slightly different phase or frequency could create a *beat pattern* that extends outward – the envelope of this beat could fall off as $1/r$, mimicking the Coulomb potential. If the carrier is nearly monochromatic with frequency ω_0, a small **amplitude perturbation** at one point produces a long-range envelope field because the medium supports propagation at (or near) ω_0. The **envelope Green's function** in a 3D elastic medium often has a $1/r$ behavior (for steady-state oscillations or static deformations), analogous to how the Poisson equation's solution yields $1/r$ for electrostatics. Thus, an *effective Coulomb field* can emerge as the envelope representing a standing pattern around a localized excitation (the "charge"). In the deterministic wave model, charges might correspond to **defects or topological oscillations** of the carrier that produce a static envelope pattern. Notably, Eshelby's theory of lattice defects in elasticity drew an analogy where an internal stress field plays the role of an electrostatic field, with a "source function" analogous to electric charge [19] [20]. By the same token, a **localized strain or oscillation** in the spacetime medium could produce a far-reaching strain pattern in the envelope, acting like a gravitational or electric potential well that affects other oscillations universally.

- **Field Propagation:** The envelope fields should propagate similarly to electromagnetic waves. In the simplest case, if the carrier is a harmonic wave at speed c (the characteristic speed of the medium), the envelope modulation travels at the *group velocity* v_g. For a nondispersive carrier $v_g \approx c$, so the envelope itself obeys a wave equation at speed c. This means **envelope pulses propagate like light signals**. The **effective light cone** in the theory is just the domain of dependence given by the medium's characteristic speed for the envelope. If the envelope equations are analogous to Maxwell's, then changes in an envelope "electric field" propagate at speed c. The deterministic wave theory can thus recover electromagnetic wave propagation as nothing more than **beat patterns** on the substrate's high-frequency oscillation. A striking example in nonlinear optics is the generation of **terahertz or DC fields by a laser**: a short, intense optical pulse (carrier) with an asymmetric envelope can produce a burst of much lower-frequency radiation (even a DC offset) – this is analogous to the envelope generating new frequencies beyond the original narrowband [21]. Likewise, the **broadband nature** of the envelope allows it to carry information (signals) and mediate forces, just as the electromagnetic field does.

In summary, envelope modulation of a narrowband carrier provides a natural mechanism for **emergent electromagnetism**: the envelope's dynamics can include static $1/r$ potentials, Coulomb-like interactions (via overlapping envelope fields), and wave propagation at effective speed c. The carrier wave is essentially a **hidden substructure** that is always present (like a high-frequency "pilot wave"), while the observable electromagnetic phenomena are encoded in the envelope. Under appropriate scaling, one can identify the envelope field with the electromagnetic 4-potential or field strengths. For instance, small deviations in phase of the carrier could correspond to vector potential components, while amplitude variations correspond to charge density, etc. Importantly, because the envelope is a *derived* quantity, it can be broadband and **apparently independent** of the single frequency of the carrier – exactly how a radio antenna's high-frequency current produces a broad spectrum of radiowaves via modulation.

4. Robust Emergent Lorentz Symmetry from a Preferred Frame

Even though the deterministic model assumes an underlying medium (a "preferred frame"), it is possible for **Lorentz invariance to emerge** as an approximate symmetry at large scales. Many condensed-matter systems exhibit this kind of emergent relativity. For example, **graphene's electrons** move on a fixed atomic lattice, yet low-energy excitations behave as massless Dirac fermions with an effective "light speed" ($\sim c/300$) and obey relativistic dispersion. The key is that the **low-energy excitations live on a single relativistic branch** of the dispersion relation. If the spacetime substrate has a stable carrier wave (as discussed in Gap 1), small perturbations (envelope excitations) might see a linear, isotropic dispersion $\omega \approx c\,k$ – defining an effective light-cone. Any **superluminal modes** or additional branches (from the medium's microphysics) would lie above some cutoff frequency or require excitation of different polarization modes, and if those are not excited (or are heavily suppressed), they do not spoil the emergent Lorentz symmetry. In essence, the medium can have "hard" degrees of freedom that only activate at high energy, while all soft excitations are constrained to **propagate at a universal speed c**. This separation of scales underpins many analogue relativity models [12] [13]. Small perturbations of a uniformly moving fluid, for instance, obey an **acoustic metric** (Unruh's sonic black hole analogy) where the speed of sound is the invariant speed. Similarly, in a pre-stressed elastic medium, linearized perturbations can see an **effective curved spacetime metric**, as shown by Cherubini & Filippi: a large static strain can produce an analog gravitational metric for sound waves [12] [13]. If the background is homogeneous, the effective metric is just Minkowski (flat) with a preferred frame at rest in the medium. But crucially, observers made of the medium's excitations cannot detect that frame if **all their experiments involve only low-energy waves** – they will get the same speed c in all directions, etc., because they themselves are composed of those waves.

To suppress superluminal leakage or multi-branch propagation, the theory can invoke several mechanisms: **gaps, damping, or kinematic constraints**. For example, if a second (faster) mode exists (say a longitudinal mode in the elastic ether with speed $c_{\ell} > c$), a localized wave-packet might try to emit that mode (analogous to Cherenkov radiation in a medium). However, if the medium is such that **energy-momentum conservation** forbids that emission at low energies, or if the faster mode is **heavy** (requires a threshold energy to excite), then it will not be excited by low-energy processes. This is akin to how in many field theories certain high-frequency modes decouple at low energy. The deterministic theory could feature a **nonlinear dispersion** where only one solution branch is populated in the regime of interest. Indeed, the success of Lorentz invariance in nature suggests that any preferred-frame effects (Lorentz-violating dispersion or interactions) are extremely small at accessible energies. This can be ensured if the elastic substrate's properties lead to Lorentz symmetry as an **infrared fixed point** – the system "flows" to a relativistic scaling at long wavelengths. Research on Einstein-Æther theories and Horava–Lifshitz gravity also deals with embedding Lorentz violation at high scale while respecting tight low-energy bounds [22] [23]. Experimentally, deviations (like energy-dependent photon speeds or symmetry violations) are bounded to very high scales, suggesting any preferred-frame substrate must produce corrections suppressed by perhaps the Planck scale or beyond [23]. The wave theory could satisfy this by having an underlying cutoff (lattice spacing or maximum frequency) that is ultra-high, so that for all practical scales the dispersion is ultra-linear and Lorentz symmetry holds to high precision.

Localized solitons as rods and clocks: In a Lorentzian theory, measuring space and time involves using objects (rods, clocks) that are ultimately made of the same fields. If solitonic wave-packets are the model's "particles," they themselves can serve as rulers and clocks. A robust emergence of Lorentz symmetry means that these solitons will experience time dilation and length contraction when moving through the medium. Remarkably, Louis de Broglie anticipated this idea: a particle is associated with an internal oscillation (de Broglie's "clock") of frequency $\nu_0 = mc^2/h$ at rest, which appears slowed down when the particle moves. In the deterministic wave model, a **classical soliton** could carry an internal beat frequency (perhaps the difference between carrier and envelope frequency) that corresponds to its rest energy. If the soliton moves, the carrier wave underpinning it must Doppler shift, causing the internal oscillation observed in lab frame to slow – thus the soliton's internal phase ticks off proper time. The soliton's shape likely also Lorentz-contracts due to the aberration of the carrier phase fronts. **Empirical support** for this concept comes from simulations of oscillatory soliton solutions: they show precisely de Broglie's double-solution structure, with a high-frequency carrier and a localized envelope (the particle) guiding it [24]. In one model, a breather-type soliton has an internal oscillation that matches what de Broglie hypothesized for the electron [24]. This provides a natural *clock*: the soliton's internal phase can represent its proper time, and two identical solitons could synchronize their phase tick rates when at rest. When one moves relative to another, **phase comparisons** would reveal time dilation. Similarly, the distance between two solitons can be gauged by sending wave signals; since all signals travel at c within the medium, the usual special relativistic structure (light-cone, simultaneity conventions) emerges for the soliton-based observers.

To avoid any "absolute" effects, the theory might also invoke a form of **Mach's principle** or dynamical substrate that hides the rest frame. For instance, if the entire universe's substrate is oscillating in the same carrier mode, an observer at rest w.r.t. the substrate vs one moving would have no local way to distinguish that beyond measuring tiny anisotropies in high-energy processes (which could be beyond current reach). The **suppression of Lorentz-violation** can thus be natural if the substrate is very stiff or the solitons carry some sort of "pocket metric" with them. Some proposals even suggest the preferred frame could be physically unobservable if all matter is made of waves that also condensate in that frame (so the frame is a hidden common reference).

In conclusion, a preferred-frame substrate can still yield **effective Lorentz invariance** for its low-energy excitations by confining those excitations to a single relativistic dispersion branch and by providing internal clocks (solitons) that transform covariantly. Any superluminal modes are assumed either *inaccessible* (require energies we can't reach) or *non-interacting* with the soliton sector (perhaps they decay rapidly or are non-propagating in vacuum). This ensures signals and particles all respect an emergent light-cone. The localized solitons themselves embody the relativity: they behave as if living in a Minkowski spacetime, using their internal oscillations as proper time and their separation defined by signal exchanges at speed c. Essentially, **Lorentz symmetry becomes an emergent symmetry** of the effective field equations, with the preferred frame only revealing itself in extreme regimes or subtle higher-order effects (much like how phonon dispersion reveals the atomic lattice only at high momenta).

5. Spinorial Holonomy and Fermionic Behavior of Solitons

It is a remarkable fact that certain **classical field solitons** can exhibit properties analogous to quantum spin-½ particles. The mathematical origin lies in the topology of the soliton's configuration space. If the space of field configurations for a single soliton has a nontrivial fundamental group π_1, then a 2π rotation of a soliton in physical space may correspond to a *non-contractible loop* in configuration space. Finkelstein and Rubinstein famously showed that one can impose a condition on the soliton's quantum wavefunctional such that encircling this loop multiplies the wavefunction by -1 [25] [26]. This is the **Finkelstein–Rubinstein (FR) constraint**, and its effect is to quantize the soliton with **half-integer spin and Fermi–Dirac statistics**. In practical terms: a full 360° rotation of the soliton does not return the state to itself, but to *minus* itself, which is exactly the behavior of a spin-$\frac{1}{2}$ object (since rotating a spinor by 360° changes its sign). Likewise, exchanging two identical solitons can be viewed as a composite rotation in the two-soliton configuration space; if this path is also topologically nontrivial (homotopic to the single-soliton 360° rotation path), the exchange of two solitons also yields a -1 phase factor [25]. Thus the multi-soliton wavefunction is antisymmetric under exchange – the hallmark of fermionic *exchange statistics*. **Theorem:** In the Chronon field theory (a specific deterministic field model), it was proven that a 2π spatial rotation of a single soliton and the exchange of two solitons both correspond to the nontrivial element of π_1 of configuration space (isomorphic to $\mathbb{Z}_2$), leading to a Berry phase of π (i.e. a -1 factor) [25] [26]. This formal result confirms that **solitons can behave as fermions**.

Concrete examples bolster this claim. In the Skyrme model of nucleons, Finkelstein & Rubinstein (1968) showed that the Skyrmion (a soliton with π_3 topology) can be quantized as a fermion by assigning a wavefunction sign change under 2π rotation. More recently, studies of **Hopf solitons** (twisted loop configurations in an $O(3)$ nonlinear sigma model) have applied the FR method: for Hopf charge Q odd, the soliton can be quantized as a fermion [27] [28]. The reasoning is that an **odd Hopf charge** soliton has a configuration space loop corresponding to a full rotation that cannot be continuously shrunk (it introduces a twist in the Hopf mapping). By requiring the wavefunction to pick up a -1 on that loop, one effectively ensures only half-integer spin states exist for the quantized soliton [27]. The outcome is that the **collective coordinate quantization** of these classical solitons yields a spin-$\tfrac{1}{2}$ doublet of states. In more general terms, whenever the order parameter or field takes values on a manifold that is doubly connected (has a $\mathbb{Z}_2$ fundamental group), soliton solutions can have **spinorial holonomy**. An intuitive picture is to consider a 3D vector field that represents some internal orientation of the medium; a 360° rotation might not bring every microscopic director back to itself (similar to how rotating a Möbius strip 360° does not return it to the same orientation).

Pauli exclusion (exchange antisymmetry) then follows naturally: if two identical soliton-fermions are in the same quantum state, swapping them multiplies the total wavefunction by -1, but being bosonic objects classically, one might have naively thought the wavefunction should be symmetric. The only way to reconcile this is if the state where they overlap is disallowed or goes to zero amplitude – effectively, they cannot both occupy the same one-particle state without the wavefunction canceling. In practice, one can say the FR sign ensures a **many-soliton wavefunction is antisymmetric** under exchange, so it automatically satisfies the Pauli exclusion principle (no two identical fermions in the same state). Thus, in a soliton model of matter, the stability of matter (pauli exclusion giving rise to electron shell structure, etc.) could emerge from these topological phase factors.

It is important to note that the FR approach is one way to get fermionic behavior, but there are also other mechanisms in field theory for spin-statistics. For instance, the π_1 loop could also produce a -1 Berry phase dynamically (as a **geometric phase**) without imposing it by hand, if one computes the fermionic determinant or zero-mode structure around the soliton. The Chronon theory reference shows that by explicitly computing the Berry phase for a rotating soliton, one indeed gets -1 [29] [30], matching the FR argument. This is a strong consistency check that a *classical* soliton, when "quantized" (or rather, when one considers small fluctuations and the corresponding quantum state), behaves as a fermion [30] [31].

In summary, **classical solitons can acquire spin-$\frac{1}{2}$ holonomy** through topological constraints. The mechanism involves the double-valued nature of the orientation (or other order parameter) configuration of the soliton. By enforcing single-valuedness of the physical state (up to sign), the soliton's 360° rotation is effectively a 720° rotation requirement for returning to the original state, exactly as for quantum spinors. The exchange of two solitons similarly corresponds to a 360° rotation in the two-particle space, giving a -1 phase [25]. This yields **fermionic exchange statistics** from a purely classical origin. Therefore, if particles in the deterministic wave theory are solitonic excitations, one can achieve **Pauli exclusion and Fermi statistics** as an emergent, topologically protected property. Notably, this does **not** require any intrinsically quantum ingredient – it is a consequence of the interplay between topology and the requirement for single-valued wavefunctions (or deterministic state variables) on configuration space.

6. Gravity-Like Long-Range Potentials in Nonlinear Elasticity

In a pre-stressed nonlinear elastic medium, one can indeed obtain long-range interactions that qualitatively resemble gravity – **universal $1/r$ potentials that couple to all forms of localized energy**. The basic intuition is that a **local stress or density perturbation curves the effective propagation geometry** for other waves. This is analogous to how in General Relativity mass-energy curves spacetime and deflects matter. In the context of elasticity: a static deformation of the medium can create a region of altered wave speed or refractive index, causing other waves (or solitons) to bend toward it. For instance, a *dent* or *tension* in a drumhead will cause waves on the drumhead to focus – similarly, a localized concentration of stress in the spacetime substrate could attract other excitations. Cherubini and collaborators showed that for a simple 1D deformation in a nonlinear solid, the small oscillations obey an effective metric equation identical to a section of Einstein's equations [12] [32]. This suggests that **an elastic medium under stress can simulate a curved spacetime** for perturbations. If we generalize to 3D, a static spherically symmetric "defect" (such as a region of higher density or pre-compression) might produce an analog of the Schwarzschild metric for the wave excitations – i.e., a gravitational well.

Mathematically, one approach is to look at the **elastic Green's function**: In linear elasticity, a point force in an infinite medium produces a displacement field that decays as $1/r$ (this is Newtonian in form). Eshelby's theory of defects treats dislocations and inclusions as sources of internal stress and finds that their far-field strain/stress decays as $1/r^2$ or $1/r$ depending on dimensionality, analogous to field lines from charges [19] [33]. In particular, Eshelby noted an analogy between an internal stress distribution and electrostatics, with an internal "source function" playing the role of charge generating a field in the continuum [20]. If we translate this to our spacetime substrate: a soliton or localized high-frequency excitation is a concentration of energy that will induce some static stress field in the surrounding medium (e.g. it might stretch or compress the medium slightly around it to satisfy boundary conditions at infinity). That static distortion could fall off as $1/r^2$ in strain (like the far-field of a local pressure), but the *potential* felt by another similar object might integrate that strain leading to a $1/r$ potential energy. In short, **localized deformations in an elastic solid create long-range elastic fields**, and any other object coupling to the same medium will feel a force. Crucially, this coupling is **universal** – it doesn't depend on the composition of the second object, only that it also interacts via elastic stresses. In the spacetime context, *everything* is made of the substrate, so everything would indeed respond. This is analogous to how gravity's curvature affects all matter and light, irrespective of their properties.

More formally, one can derive an emergent Poisson equation for the envelope of the substrate. In some approaches (e.g. Sakharov's induced gravity or analog gravity), the **Einstein field equations** or Newton's law appear as an emergent equation for the background. The Chronon field theory cited indicates that the coupling of the chronon field (their spacetime field) produces an Einstein-like term in the effective action, and Newton's constant G emerges from the material parameters and cutoff [34] [35]. Essentially, the stiffness of the medium and the scale of dispersions determine the "strength" of the induced curvature (much like stress-strain relation: softer medium means more curvature per unit energy). If the deterministic wave model has a term analogous to $\gamma R_{\mu\nu} u^\mu u^\nu$ (as Chronon theory does [34]), expanding that yields an effective Einstein-Hilbert action with a coefficient $\propto 1/G$ [34] [35]. That indicates gravity is not fundamental but an emergent phenomenon from elastic couplings, with G determined by the coupling strength and cutoff (so it "can be derived, not inserted" [36]).

From a more physical viewpoint: imagine the substrate has an equilibrium state, and you introduce a heavy oscillating concentration (like a "particle" soliton). The substrate might sag or stretch around it – for instance, if the medium has an effective mass density, that concentration creates a potential well.

Other particles (solitons) will tend to drift toward that well because moving into a region of stretched substrate could lower their energy (just as masses move toward lower gravitational potential). The substrate's prestress is key – if the medium is extremely stiff, the deformation will be tiny (hence a weak potential), whereas a softer medium yields more pronounced curvature from the same object (stronger gravity). The **universality** is ensured because all particles are just patterns in this medium, so they all follow the geodesics of the deformed medium. In analogy, photons follow curvature just as planets do, in General Relativity; here, a wave-packet (photon analog) and a soliton (matter analog) both travel through the same distorted elastic medium and hence both bend in the same way.

One concrete mechanism in elasticity that yields a $1/r$ potential is the presence of a **pre-stress or pre-strain** that couples to incremental deformations. If the medium has a resting stress (like tension in a drumhead), adding an extra local stress will send out a static perturbation that decays slowly. In fact, gravity could be modeled by something like a defect that induces a strain proportional to its "mass" and inversely with distance – similar to how inserting a mass in a stressed membrane causes a dip. There is precedent in *analogue gravity*: a tensile membrane with a mass on it has an identical equation to Newtonian gravity in 2D (this is sometimes used as a visualization of GR). More rigorously, in **analogue gravity frameworks**, one often finds the emergent gravitational potential is encoded in the **index of refraction** seen by perturbations. For instance, a flowing fluid with a non-uniform velocity can mimic a gravitational field – all excitations see a curved space-time metric (the acoustic metric) [12] . By designing the flow or stress field, one can simulate gravitational phenomena such as **horizons or light bending**.

In the context of this deterministic wave theory, one could incorporate gravity-like effects by allowing the **cosmological background oscillations** (the carrier) to influence the effective stiffness of the medium. The preprint suggests that persistent sub-Planckian oscillations leave a nearly uniform offset in the Einstein-like equations, acting as a cosmological constant [37] . Likewise, spatial variations in those persistent oscillations could act like mass. The effective gravitational potential would then be encoded in a slowly varying part of the stiffness or strain field $\Delta E(x)$ in their notation [38] . Solitons – being concentrations of energy – would modify $\Delta E(x)$ and in turn create an *attraction* for other solitons (since it effectively alters their dispersion relation locally to lower energy when closer). In short, if one **coarse-grains the elastic medium**, one can obtain an effective field (call it $\Phi(x)$) that plays the role of the gravitational potential. All matter waves and particles follow Φ via something like $L \sim \rho |\nabla \Phi|^2 + \Phi T^{00}$, leading to equations similar to Poisson's equation $\nabla^2 \Phi \propto T^{00}$.

To summarize, by **pre-stressing or dynamically deforming** the nonlinear elastic substrate, one introduces a field that couples to all excitations identically (since it's literally the metric or background in which they move). This yields gravity-like inverse-square forces and even the full machinery of curved spacetime for the emergent fields. Known models and analogies support this: from continuum defect theory (stress functions analogous to charges) [20] , to induced gravity (where G is derived from microphysics) [35] , to analog gravity in fluids and solids (where sound sees a metric) [12] . The challenge for the deterministic wave theory would be to get not just a **Newtonian 1/r potential** but the **full Einsteinian structure** (with frame-dragging, etc.). However, since Lorentz symmetry can emerge (Gap 4) and gauge fields can emerge (Gap 2), having the substrate's stress behave like a spin-2 field (graviton) is conceivable. In fact, small ripples of the background stress distribution would be an **elastic wave mode** that likely corresponds to a spin-2 graviton in the effective theory. The **universality** of gravity in this picture is automatic: everything is made of the substrate, so everything is affected by its shape – which is exactly the principle of equivalence realized in a mechanical sense.

7. Quantum-Like Statistics from Classical Soliton Dynamics

One of the most intriguing challenges is how to obtain **quantum statistical behavior** – randomness, discrete outcomes, interference – from a deterministic classical system. Recent research indicates this is possible through mechanisms like **deterministic chaos, threshold detection, and phase averaging**. In a classical chaotic system, a lack of precise knowledge (or practical uncontrollability) of initial conditions can yield observational results indistinguishable from intrinsic randomness. The deterministic wave theory can harness this by positing that the underlying soliton dynamics are chaotic on very fine scales (e.g. sub-Planckian scales), producing outcomes that *appear* random to coarse-grained observers. In fact, a theorem by Coles *et al.* (2021) showed that entanglement with a chaotic environment in pilot-wave models drives any initial distribution to the Born $|\Psi|^2$ law, analogous to an H-theorem for quantum equilibrium [39]. Essentially, **deterministic chaos leads to a fast relaxation** of particle configuration distributions to the $|\psi|^2$ distribution [39]. This supports the idea that the Born rule could emerge from ergodic exploration of phase space by the hidden variables.

Discrete events and the Born rule can also come from the combination of continuous waves and **threshold detectors**. Real measuring devices (photographic plates, Geiger counters, etc.) respond only when a certain energy or intensity threshold is exceeded. In a classical field picture, if a field (e.g. an electromagnetic wave or our substrate wave envelope) is below threshold, no click is recorded; once it builds up enough energy in a localized region (by interference or fluctuation), a discrete detection event ("quantum" particle hit) occurs. This way, a continuous wave can yield discrete outcomes. Several works (Khrennikov *et al.*, 2012; as well as the **Prequantum Classical Statistical Field Theory** program) have shown that if you model detectors as threshold devices, the statistics of detection events follow the Born rule even though the underlying field is continuous [40] [41]. In short, the **counting of threshold-crossing events** by a continuous random field yields $P \propto |\text{field}|^2$. In our context, the soliton wavefunction intensity $|\Psi|^2$ (which is actually a classical energy density in the substrate) would dictate the frequency of detector clicks. The randomness arises because the exact moments and positions where the threshold is crossed are sensitive to microscopic details (chaos), so to an experimenter they appear random, with only the probability distribution being predictable. This is exactly how quantum experiments behave.

We also have experimental analogs: the famous **oil droplet experiments** by Couder & Fort showed that a bouncing droplet on a vibrating fluid bath (a deterministic pilot-wave system) can mimic single-particle quantum behaviors like double-slit interference, quantized orbits, and tunneling. The droplet's motion is chaotic but with an underlying pilot wave guiding it, and statistical outcomes (like interference fringes in a double-slit analog) match the quantum predictions when averaged over many trials. Very recently, Ceausu & Dagan (2025) demonstrated that particles sliding on a shallow water wave (again a classical system) exhibit **quantum-like statistical outcomes** [42] [43]. They found that the spatial distribution of the particles matches the $|\psi|^2$ pattern for a quantum particle in the same potential [43]. The system showed both periodic (predictable) and chaotic trajectories, and remarkably the *ensemble* of chaotic ones reproduced the probabilistic quantum result [44] [43]. In their words, "quantum-like statistics can emerge from deterministic dynamics, with intermittent chaos and order playing a central role," and the probability of finding the particle corresponded to the wave intensity squared [44] [43]. This provides strong evidence that a classical wave/particle system can yield an **apparent Born rule**. Moreover, they noted that this suggests what looks like intrinsic randomness in quantum mechanics may in fact arise from "structured but unpredictable motion" in a nonlinear classical system [45].

Another aspect is **interference**. In a deterministic wave model, interference is natural: the waves always interfere, creating patterns. If particles (solitons) ride on these waves or interact with them (e.g. via phase coupling), they will exhibit interference fringes in their motion probabilities. The double-slit experiment, for instance, would be described by the carrier wave passing through both slits and forming an interference pattern in the envelope; a soliton or wave-packet passing through will be nudged by this pattern, resulting in detection statistics that show fringes. If each detection is a threshold event, individual impacts still build up the interference pattern frequency distribution. Thus the *observables* show interference even though each particle took a definite (but chaotic) path. This is essentially a **de Broglie–Bohm pilot wave** scenario executed with classical waves. The deterministic wave theory appears to incorporate this by having the particle as a localized high-frequency excitation and the pilot wave as its extended low-frequency component; collapse or decoherence occurs via nonlinear damping terms (the STM model, for example, uses non-Markovian damping to mimic wavefunction collapse) [46] [47].

Finally, **discreteness of quanta** can emerge if solitons or wave modes come in quantized units. Classical nonlinear wave equations often have discrete soliton solutions characterized by an integer topological charge or quantized energy. These could correspond to “quantum” particles with discrete energies. Additionally, when energy is exchanged via waves in a bounded system, normal mode frequencies lead to quantized energy levels (like a vibrating string only supports certain frequencies). In a cosmological setting, if the whole universe’s carrier mode is like a large resonator, maybe only certain global frequencies (energies) are allowed. But even without that, detectors respond in discrete lumps because they register whole solitons or whole wave quanta at a time (you can’t get half a click – this is an all-or-nothing response). This effectively digitizes the energy.

In summary, **quantum-like statistics** in a deterministic soliton system can be explained by a combination of chaotic dynamics and nonlinear measurement effects. The **Born rule** ($P=|\psi|^2$) likely arises from ergodicity or typicality: after decoherence-like interactions, the system’s hidden variables get distributed according to $|\psi|^2$ (as seen in pilot-wave chaos studies [39] and analog experiments [43]). **Discrete events** come from threshold-triggered detection or quantized soliton number (each soliton carries a fixed energy, like one “quantum”). **Interference** emerges naturally from wave superposition, and because the particles are coupled to waves, their outcomes reflect those interference patterns even though each particle individually had a deterministic trajectory. Thus, a classical wave framework can reproduce all the puzzling quantum statistics without true randomness – randomness is only apparent, stemming from our ignorance of the complex underlying state (which might be practically unknowable if it’s chaotic on tiny scales). The literature supports this view: “what appears as intrinsic randomness may instead arise from structured but unpredictable motion in nonlinear systems” [45].

8. Mapping Theory Parameters to Physical Constants (c, ℏ, e, G)

For the deterministic wave theory to be viable, its parameters (such as the density and elasticity of the substrate, characteristic lengths, etc.) must be chosen so that the *emergent* constants – speed of light c, Planck’s constant $\hbar$, elementary charge e, gravitational constant G – take their known values, all while avoiding physical contradictions (like an absurd energy density of the vacuum, or violations of experimental bounds). Achieving this is a delicate calibration problem: essentially one must identify the correspondence between the model’s fundamental scales and the SI units of these constants.

- **Speed of light (c):** In this theory, c is presumably the propagation speed of transverse waves in the spacetime medium (the carrier or envelope speed). For an elastic medium, wave speed is $c = \sqrt{K/\rho}$ (for some effective stiffness K and density ρ). To get $c \approx 3\times10^8$ m/s, one can adjust K and ρ. If ρ were like a normal material density (say on order 1000 kg/m^3), K would need to be enormous ($\sim 10^{16}$ N/m^2) to yield that c. In fact, plugging numbers: $c^2 \sim 9\times10^{16}$, so $K/\rho \sim 9\times10^{16}$. With $\rho\sim10^3$, $K \sim 9\times10^{19}$ Pa. That is far beyond ordinary materials (steel is 10^{11} Pa) but not reaching Planck scales yet. The preprint “Refined STM” hints that a required stiffness on the order of 10^{31} Pa might be needed if not using Planck-scale moduli [48]. 10^{31} Pa corresponds to an energy density of 10^{30} J/m^3, which is extremely high – about 10^{13} times nuclear density. If such a high energy density filled the vacuum, naive gravity would curl up the universe (the cosmological constant problem). The way out is that this energy might be *mostly uniform* and only gradients matter. A perfectly uniform energy density (of any size) does not gravitate in Newtonian terms and in GR acts like a cosmological constant. The deterministic wave theory might leverage this: the huge base energy of the substrate is interpreted as an effective cosmological constant (or cancelled by a pressure term), leaving only small perturbations as sources of gravity [37]. In other words, the “rest energy” of the medium could be hidden as dark energy, with only deviations contributing to local gravity. This avoids immediate conflict with astronomical observations, provided the net cosmological constant from this is within measured bounds (which are around 6×10^{-27} kg/m^3, vastly smaller – suggesting some cancellation mechanism is needed if the bare density is large).

- **Planck’s constant ($\hbar$):** In the wave model, $\hbar$ is related to the scale of action of emergent quantum fields. One way to identify $\hbar$ is to look at the effective Schrödinger equation for the envelope: $i\hbar \partial_t \psi = -(\hbar^2/2m)\nabla^2\psi + \cdots$. By deriving this from the underlying parameters, one can read off $\hbar$. For example, in the STM model, a coarse-graining yields an effective $\hbar_{\text{eff}}$ that depends on the energy and time scales of the carrier oscillation [36]. The Chronon field theory states $\hbar_{\text{eff}} \sim E_c \tau_c$, where E_c and τ_c are characteristic energy and time scales of the chronon (basically the internal oscillator’s energy and period) [36]. If our substrate has a natural frequency ω_0 and some characteristic length ℓ_0, we might get $\hbar \sim \rho \omega_0 \ell_0^3$ (just dimensional analysis for action). To match $\hbar = 1.05\times10^{-34}$ J·s, at the scale of, say, $\ell_0 \sim 10^{-15}$ m (typical nuclear scale) and $\omega_0 \sim 10^{22}$ s^{-1} (Compton frequency for electron), ρ would need to be tuned so that the action of one “quantum” of the carrier equals $\hbar$. There might be some topological invariant that enforces quantization of action in units of $\hbar$. For instance, in analogies like circulation in superfluids, the circulation is quantized in units h/m. If the spacetime medium has something analogous (maybe a minimum increment of phase slip corresponds to $\hbar$ action), then $\hbar$ can be seen as a derived unit. Another viewpoint is that **$\hbar$ sets the scale of zero-point fluctuations**. In stochastic electrodynamics (a classical approach), one assumes a classical zero-point field with spectral density $(\hbar\omega/2)$ per mode to recover quantum behavior. The presence of $\hbar$ might signal the amplitude of intrinsic microscopic oscillations. If the carrier waves of frequency ω_0 have a tiny amplitude such that each mode of volume $\sim \lambda^3$ carries energy $\hbar\omega_0$, that builds in $\hbar$. The deterministic theory might allow calculation of $\hbar$ from the medium’s micro-parameters; for example, Chronon theory says these constants “are not postulated but appear as emergent normalizations” in the long-wavelength limit [49].

- **Elementary charge (e):** The coupling constant e (for electromagnetism) would emerge in the effective gauge field action. If the gauge field is coming from Berry phases (Gap 2) or from some effective current in the medium, e will be related to how the envelope (matter wave) interacts with the emergent gauge potential. In a normal gauge theory, the minimal coupling is $e A_\mu$. In our scenario, A_μ is derived from medium variations, and the “charge” e would be proportional to the degree to which a given excitation’s phase responds to those variations. If, for instance, the polarization basis rotates by an angle when moving a certain distance in a modulated medium, the phase change is geometric. To mimic an electron's coupling, one calibrates that such a rotation over a loop of certain field strength yields a 2π phase (the Aharonov-Bohm condition). The **fine structure constant** $\alpha = e^2/4\pi\hbar c \approx 1/137$ would be a natural target: does the theory yield a dimensionless combination of parameters that equals 1/137? Perhaps the ratio of the carrier energy density to some stiffness, or a topological coupling number. In Chronon theory, they claim e emerges as a normalization constant in the long-wavelength action [49]. This implies that e could come from matching the strength of the emergent $U(1)$ gauge field to the observed Coulomb force. If the analog Coulomb potential is $\Phi(r) \sim \frac{Q_{\text{eff}}}{4\pi \epsilon r}$ in the medium (with some effective permittivity ϵ set by medium properties), then identifying Q_{eff} for a unit topological charge soliton with the electron's charge e gives a handle. Possibly, Q_{eff} is quantized by the topology of the medium (like how Dirac quantization can come from topology in a gauge field). Thus e might be fixed by a requirement that a certain **phase change around a loop = 2π** corresponds to one unit of electric charge. In a Berry phase picture, the total phase around a monopole in parameter space is 2π, relating to quantization of charge. So it's plausible the medium's geometry enforces charge quantization and sets the value of e. Ultimately, fitting e likely involves matching the model's **field strength of a source** to Coulomb's law at a reference distance, which in turn constrains a combination of substrate parameters (like anisotropy gradient per unit energy, etc.).

- **Gravitational constant (G):** If gravity emerges from the elasticity of the medium (as per Gap 6), then G is determined by how much curvature (or attractive potential) is produced by a given amount of energy. In a mechanical analog, if I have a mass M on a tense membrane, how deep is the dent? G would relate to that depth per unit M. Chronon theory results indicate that G is induced by either direct coupling or radiative corrections and is proportional to the inverse of some combination of coupling constants times cutoff (essentially $G^{-1} \sim c_s \Lambda^2$ in their formula) [36] [35]. This is reminiscent of Sakharov's induced gravity where $G^{-1} \sim \Lambda^2$ (cutoff squared) divided by a factor from quantum degrees of freedom. In a classical elasticity context, one can derive the **elastic analog of Poisson's equation**: $\nabla^2 \Phi = \kappa \rho_{\text{mass}}$, where κ would play the role of $4\pi G$ in Newton's law. This κ would be a function of the medium's compressibility and density. If the medium is extremely stiff, κ is very small (gravity is weak). So there is a trade-off: making c large by high stiffness also tends to make G small (since the medium hardly curves under mass). The known values c and G are such that the **Planck impedance** ($\sqrt{G}/c$) is an incredibly tiny number (around 10^{-18} in SI units), hinting that indeed the medium is both very stiff and high-density (so that c is high but G is low). The theory's parameters must be tuned to get this right. Possibly, one chooses the cutoff scale (maybe around Planck length $\sim10^{-35}$ m) where new physics comes in. If the fundamental length of the medium is the Planck length, then naturally one expects the gravitational coupling to come out correctly (since that's how Planck units are defined). But using Planck-scale lattice or moduli can lead to huge vacuum energy. The preprint suggests that without Planck-scale moduli, they can still get the right order of magnitudes by multi-scale effects [48]. Perhaps the carrier oscillations (sub-Planckian) effectively renormalize the stiffness, yielding an **effective gravitational coupling** that matches observations while the bare parameters don't blow up energy density. For example, a **scale-dependent stiffness** could mean the medium is softer at cosmological scales than microscopically (providing more curvature for a given energy on large scales, which could mimic G). This is speculation, but the idea is that **G emerges from the interplay of micro and macro elasticity**.

- **Avoiding unphysical densities:** One major concern is avoiding contradictions with experiments like the absence of dispersion or energy loss in high-frequency phenomena. If the medium's cutoff frequency or lattice spacing is too low (say at 10^{-20} m or something), high-energy cosmic rays might probe it and reveal Lorentz violation, which hasn't been seen. So likely the fundamental scale has to be Planckian (~10^{-35} m, 10^{43} Hz) or at least very high. That in turn often implies Planck-density energy in the vacuum. To not have such energy curve spacetime drastically, one may need to invoke a **balancing mechanism**: perhaps the medium has tension as well as energy such that its stress-energy nearly cancels (similar to how a cosmological constant can be viewed as positive energy density plus negative pressure). If the medium's equation of state is exotic (like stiff or with w ≈ -1), it could be present without affecting local physics except through small gradients. The **Higgs field vacuum expectation** in the Standard Model is an example: it's huge in energy density ($\sim10^8$ J/m^3), but we don't gravitate towards it because it's uniform and possibly canceled by other terms. The wave theory could posit that the **vacuum is a resonant state** of the substrate with zero *net* stress (or an incompressible medium where uniform energy doesn't gravitate). Then only variations in energy (which are much smaller) act as sources of gravity – matching reality.

As a concrete mapping: the theory might assign, say, substrate density ρ_0, bulk modulus B_0, and a small scale cutoff a_0. One finds $c = \sqrt{B_0/\rho_0}$, $\hbar = \rho_0 a_0^5 c$ (just a guess on how a quantum of action might scale), G relates to $1/B_0$ times some geometric factor, and e^2 arises from defect topology (maybe $e^2 \sim B_0 a_0^2$ in some units). The goal would be to pick ρ_0, B_0, a_0 such that these combinations hit the known constants. If a_0 is the Planck length (1.6×10^{-35} m), one naturally gets the Planck energy density scale in B_0. One could lower B_0 and ρ_0 together to keep c constant but reduce energy density – at the cost of making a_0 larger (meaning new physics at a larger length). But that might conflict with precision tests of Lorentz symmetry.

The **good news** is that some independent pieces are falling into place: Chronon/chronon and STM models indicate that these constants can emerge rather than be inputs [49] [35]. They specifically note that G, e, and c *appear* as normalizations of the long-wavelength action, not put in by hand [49]. This means, in principle, once the model is fully specified, one could solve for those constants. The hope is that there exists at least one self-consistent solution that matches reality without glaring issues. The energy density issue might be resolved if the vacuum oscillations act as a kind of **reference frame that is unobservable** except through gravity – which they then interpret as dark energy. The STM paper claims the residual vacuum energy of the persistent waves is a uniform offset that acts like dark energy [37]. Notably, the observed dark energy density (~6×10^{-27} kg/m^3) is 120 orders of magnitude below naive Planck density – a huge discrepancy. They might aim to explain this by some averaging or adaptive stiffness (scale-dependent elasticity) that effectively makes the large energy non-gravitating or slowly varying [37].

In summary, mapping deterministic wave theory parameters to observable constants is challenging but feasible: one treats c as a given (sets a relation between stiffness and density), $\hbar$ emerges from the smallest scale oscillation units (and can be tuned by choosing that scale and the amplitude of oscillations such that one quantum of oscillation carries action h), e would come from the coupling strength of the emergent $U(1)$ gauge field (possibly topologically quantized), and G comes from how easily the medium's geometry bends under energy (so inversely proportional to stiffness, likely). By adjusting these, one seeks a self-consistent set that matches known values **without** producing obvious contradictions. The requirement of very high energy densities can be circumvented if that energy is uniformly distributed or self-canceling (as vacuum energy seems to be). Experimental bounds (like PPN tests, Michelson-Morley, etc.) impose that if there is any preferred-frame effect, it must be extremely small [22]. That implies the model's Lorentz restoration must be accurate up to at least 10^{-15} or better, constraining the ratio of any other mode speeds to c, etc. Ensuring this while fitting constants will probably push the model toward a Planckian fundamental scale – which is fine as long as it's coherent. The next steps would involve computing these emergent constants from the theory's equations explicitly and checking, for example, that no "unphysical" predictions (like energy non-conservation or huge vacuum Cherenkov losses) arise. So far, though, proponents of such models have shown it's at least *plausible* to get the known constants: *e.g.*, in one approach G turned out proportional to the chronon coupling and cutoff squared [35], and could be matched; e and $\hbar$ likewise were tied to coupling constants and mode frequencies that can be chosen.

Overall, **yes, the theory's parameters can be tuned to known constants**, but doing so likely implies a physically extreme substrate (incredibly high frequency and modulus) which fortunately remains hidden from direct detection. The hope is that all those "extreme" aspects manifest only in the constants of nature and perhaps in cosmological phenomena (like a small dark energy, trace Lorentz violations near the cutoff, etc.) but not in everyday physics – thus no contradictions with current experimental bounds. The literature on induced gravity and analog models provides guidance for making these identifications in a mathematically sound way [34] [35], and future work will no doubt refine the exact mapping from elasticity constants to $(c,\hbar,e,G)$ in the deterministic wave framework.

Sources: The above proposals reference established analogies and models: e.g., driven condensates and lasers for global oscillation [1]; Berry-phase gauge fields in photonic media [3] [9]; multi-scale envelope derivations [16]; analog gravity in elasticity [12]; topological quantization of solitons as fermions [25] [27]; and classical systems reproducing quantum statistics [43] [45]. These lend concrete support to each gap's resolution within a deterministic wave paradigm.

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